



MONASH University

Developing a grammar to define interactive graphics with application to complex model diagnostics

Piyadi Gamage Jayani Lakshika

B.Sc. (Hons) in Statistics, University of Sri Jayewardenepura (USJ), Sri Lanka

A thesis submitted for the degree of
Doctor of Philosophy
at Monash University in 2025
Department of Econometrics & Business Statistics

Table of contents

Copyright notice	iv
Abstract	1
Declaration	2
Acknowledgements	4
1 Introduction	0
1.1 Quarto	0
1.2 Data	0
1.3 Figures	1
1.4 Results from analyses	2
1.5 Tables	2
2 Visualising how non-linear dimension reduction warps your data	3
2.1 Exponential smoothing	3
3 Large-scale user study to explore perception and misperception in NLDR representations	5
3.1 Exponential smoothing	5
4 Generating interesting high-dimensional data structures	7
4.1 Exponential smoothing	7
5 Conclusion and future plans	9
5.1 Contributions	9
5.2 Future work	9
Bibliography	10

Copyright notice

Produced on 16 March 2025.

© Piyadi Gamage Jayani Lakshika (2025).

Abstract

Declaration

I hereby declare that this thesis contains no material which has been accepted for the award of any other degree or diploma at any university or equivalent institution and that, to the best of my knowledge and belief, this thesis contains no material previously published or written by another person, except where due reference is made in the text of the thesis.

The inclusion of co-authors reflects the fact that the work came from active collaboration between researchers and acknowledges input into team-based research. In the case of Chapter 2, my contribution to the work involved the following:

Chapter	Publication title	Status	Student contribution	Co-authors contribution	Coauthors are Monash students
2		Published in the Journal of Computational and Graphical Statistics	80% Concept, Analysis, Software, Writing		No

Chapters 3, *k*, and Chapter 4, **, are planned for submission to peer-reviewed journals.

To ensure the clarity and coherence of the written content, artificial intelligence tools were employed to assist in smoothing and refining the language throughout the thesis.

I have not renumbered sections of submitted or published papers in order to generate a consistent presentation within the thesis.

Student name: Piyadi Gamage Jayani Lakshika

Student signature:

Date: 15th January 2025

I hereby certify that the above declaration correctly reflects the nature and extent of the student's and co-authors' contributions to this work. In instances where I am not the responsible author I have consulted with the responsible author to agree on the respective contributions of the authors.

Main Supervisor name: Dianne Cook

Main Supervisor signature:

Date: 15th January 2025

Acknowledgements

This thesis would have never been a success without the guidance, encouragement, and support of my supervisors, family, and friends. This note is to show my gratitude to them.

Last, but by no means least, I would like to thank my family members. I am truly grateful to my father, P. G. K. Jagath Premalal, and my mother K. G. N. Chandima Dias for their unconditional selfless love, care, and sacrifices that they made to make my life better. Your inspiration and guidance always kept me motivated to perceive better achievements in my life. I am fortunate to be your daughter. I would also like to thank my sister Jayasha, my brother Janith, and my sister-in-law, Agashi for their unconditional love, understanding, and moral support given for completing my research.

Chapter 1

Introduction

This is where you introduce the main ideas of your thesis, and an overview of the context and background.

In a PhD, Chapter 2 would normally contain a literature review. Typically, Chapters 3–5 would contain your own contributions. Think of each of these as potential papers to be submitted to journals. Finally, Chapter 6 provides some concluding remarks, discussion, ideas for future research, and so on. Appendixes can contain additional material that don't fit into any chapters, but that you want to put on record. For example, additional tables, output, etc.

1.1 Quarto

In this template, the rest of the chapter shows how to use quarto. The big advantage of using quarto is that it allows you to include your R or Python code directly into your thesis, to ensure there are no errors in copying and pasting, and that everything is reproducible. It also helps you stay better organized.

For details on using Quarto, see <http://quarto.org>.

1.2 Data

Included in this template is a file called `sales.csv`. This contains quarterly data on Sales and Advertising budget for a small company over the period 1981–2005. It also contains the GDP (gross domestic product) over the same period. All series have been adjusted for inflation. We can load in this data set using the following code:

```
sales <- readr::read_csv(here::here("data/sales.csv")) |>
  rename(Quarter = `...1`) |>
  mutate(
    Quarter = as.Date(paste0("01-", Quarter), "%d-%b-%y"),
    Quarter = yearquarter(Quarter)
  ) |>
  as_tsibble(index = Quarter)
```

Any data you use in your thesis can go into the data directory. The data should be in exactly the format you obtained it. Do no editing or manipulation of the data prior to including it in the data directory. Any data munging should be scripted and form part of your thesis files (possibly hidden in the output).

1.3 Figures

Figure 1.1 shows time plots of the data we just loaded. Notice how figure captions and references work. Chunk names can be used as figure labels with `fig-` prefixed. Never manually type figure numbers, as they can change when you add or delete figures. This way, the figure numbering is always correct.

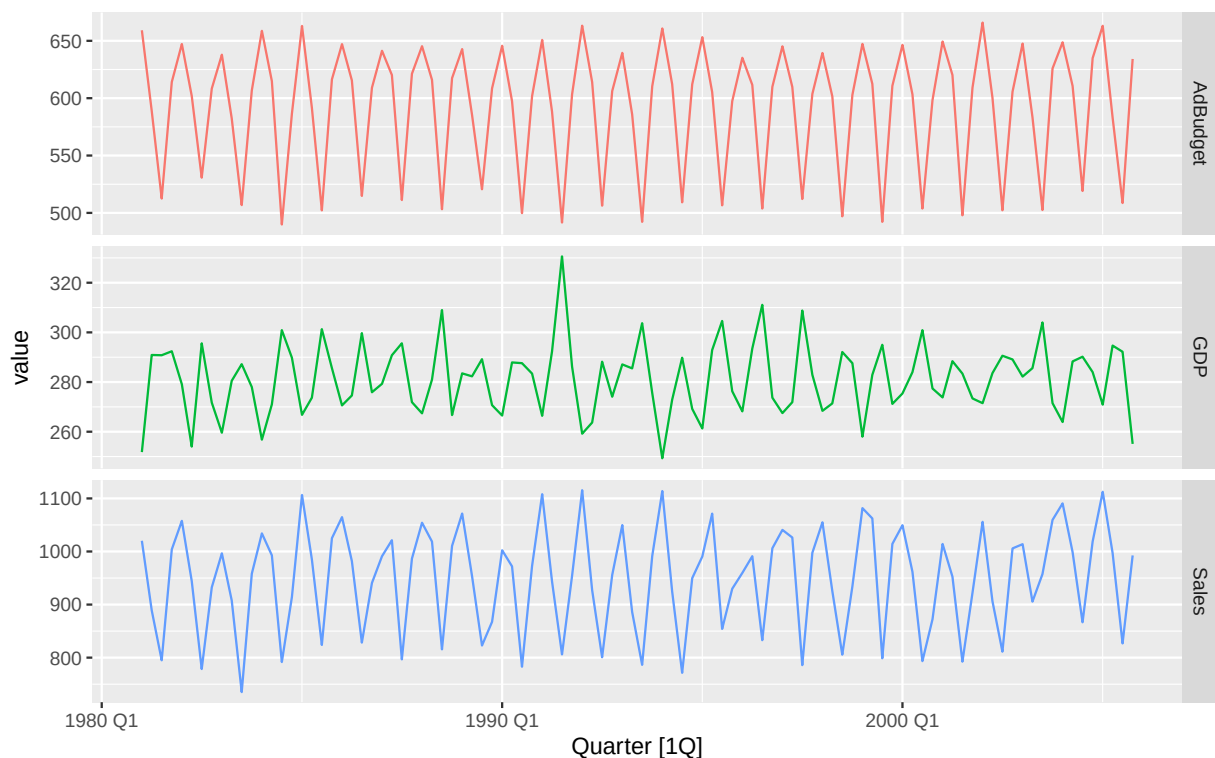


Figure 1.1: Quarterly sales, advertising and GDP data.

1.4 Results from analyses

We can fit a regression model to the sales data.

If y_t denotes the sales in quarter t , x_t denotes the corresponding advertising budget and z_t denotes the GDP, then the resulting model is:

$$y_t = \beta x_t + \gamma z_t + \varepsilon_t \quad (1.1)$$

where $\hat{\beta} = 1.85$, and $\hat{\gamma} = 1.04$. We can reference this equation using Equation 1.1.

1.5 Tables

We can also make a nice summary table of the coefficients, as shown in Table 1.1

Table 1.1: *Coefficients from the fitted model.*

Coefficient	Estimate	P value
(Intercept)	-438.98	0.02
GDP	1.04	0.02
AdBudget	1.85	0.00

Again, notice the use of labels and references to automatically generate table numbers.

Chapter 2

Visualising how non-linear dimension reduction warps your data

This chapter contains a summary of the context in which your research is set.

Imagine you are writing for your fellow PhD students. Topics that are well-known to them do not have to be included here. But things that they may not know about should be included.

Resist the temptation to discuss everything you've read in the last few years. And you are not writing a textbook either. This chapter is meant to provide the background necessary to understand the material in subsequent chapters. Stick to that.

You will need to organize the literature review around themes, and within each theme provide a story explaining the development of ideas to date. In each theme, you should get to the point where your ideas will fit in. But leave your ideas to later chapters. This way it is clear what has been done beforehand, and what new contributions you are making to the research field.

All citations should be done using markdown notation as shown below. This way, your bibliography will be compiled automatically and correctly.

2.1 Exponential smoothing

Exponential smoothing methods were originally developed in the late 1950s (Brown 1959, 1963; Holt 1957; Winters 1960). Because of their computational simplicity and interpretability, they became widely used in practice.

Empirical studies by Makridakis and Hibon (1979) and Makridakis et al. (1982) found little difference in forecast accuracy between exponential smoothing and ARIMA models. This made the family of exponential smoothing procedures an attractive proposition (see Chatfield et al. 2001).

The methods were less popular in academic circles until Ord et al. (1997) introduced a state space formulation of some of the methods, which was extended in Hyndman et al. (2002) to cover the full range of exponential smoothing methods.

Chapter 3

Large-scale user study to explore perception and misperception in NLDR representations

This chapter contains a summary of the context in which your research is set.

Imagine you are writing for your fellow PhD students. Topics that are well-known to them do not have to be included here. But things that they may not know about should be included.

Resist the temptation to discuss everything you've read in the last few years. And you are not writing a textbook either. This chapter is meant to provide the background necessary to understand the material in subsequent chapters. Stick to that.

You will need to organize the literature review around themes, and within each theme provide a story explaining the development of ideas to date. In each theme, you should get to the point where your ideas will fit in. But leave your ideas to later chapters. This way it is clear what has been done beforehand, and what new contributions you are making to the research field.

All citations should be done using markdown notation as shown below. This way, your bibliography will be compiled automatically and correctly.

3.1 Exponential smoothing

Exponential smoothing methods were originally developed in the late 1950s (Brown 1959, 1963; Holt 1957; Winters 1960). Because of their computational simplicity and interpretability, they became

widely used in practice.

Empirical studies by Makridakis and Hibon (1979) and Makridakis et al. (1982) found little difference in forecast accuracy between exponential smoothing and ARIMA models. This made the family of exponential smoothing procedures an attractive proposition (see Chatfield et al. 2001).

The methods were less popular in academic circles until Ord et al. (1997) introduced a state space formulation of some of the methods, which was extended in Hyndman et al. (2002) to cover the full range of exponential smoothing methods.

Chapter 4

Generating interesting high-dimensional data structures

This chapter contains a summary of the context in which your research is set.

Imagine you are writing for your fellow PhD students. Topics that are well-known to them do not have to be included here. But things that they may not know about should be included.

Resist the temptation to discuss everything you've read in the last few years. And you are not writing a textbook either. This chapter is meant to provide the background necessary to understand the material in subsequent chapters. Stick to that.

You will need to organize the literature review around themes, and within each theme provide a story explaining the development of ideas to date. In each theme, you should get to the point where your ideas will fit in. But leave your ideas to later chapters. This way it is clear what has been done beforehand, and what new contributions you are making to the research field.

All citations should be done using markdown notation as shown below. This way, your bibliography will be compiled automatically and correctly.

4.1 Exponential smoothing

Exponential smoothing methods were originally developed in the late 1950s (Brown 1959, 1963; Holt 1957; Winters 1960). Because of their computational simplicity and interpretability, they became widely used in practice.

Empirical studies by Makridakis and Hibon (1979) and Makridakis et al. (1982) found little difference in forecast accuracy between exponential smoothing and ARIMA models. This made the family of exponential smoothing procedures an attractive proposition (see Chatfield et al. 2001).

The methods were less popular in academic circles until Ord et al. (1997) introduced a state space formulation of some of the methods, which was extended in Hyndman et al. (2002) to cover the full range of exponential smoothing methods.

Chapter 5

Conclusion and future plans

5.1 Contributions

5.2 Future work

Bibliography

Brown, R. G. (1959), *Statistical forecasting for inventory control*, McGraw-Hill, New York.

Brown, R. G. (1963), *Smoothing, forecasting and prediction of discrete time series*, Englewood Cliffs, New Jersey: Prentice Hall.

Chatfield, C., Koehler, A. B., Ord, J. K., and Snyder, R. D. (2001), "A new look at models for exponential smoothing," *The Statistician*, 50, 147–159.

Holt, C. E. (1957), *Forecasting trends and seasonals by exponentially weighted averages*, O.N.R. Memorandum, Carnegie Institute of Technology.

Hyndman, R. J., Koehler, A. B., Snyder, R. D., and Grose, S. (2002), "A state space framework for automatic forecasting using exponential smoothing methods," *International Journal of Forecasting*, 18, 439–454.

Makridakis, S., Anderson, A., Carbone, R., Fildes, R., Hibon, M., Newton, R. L. J., Parzen, E., and Winkler, R. (1982), "The accuracy of extrapolation (time series) methods: Results of a forecasting competition," *Journal of Forecasting*, 1, 111–153.

Makridakis, S., and Hibon, M. (1979), "Accuracy of forecasting: An empirical investigation (with discussion)," *Journal of Royal Statistical Society (A)*, 142, 97–145.

Ord, J. K., Koehler, A. B., and Snyder, R. D. (1997), "Estimation and prediction for a class of dynamic nonlinear statistical models," *Journal of American Statistical Association*, 92, 1621–1629.

Winters, P. R. (1960), "Forecasting sales by exponentially weighted moving averages," *Management Science*, 6, 324–342.